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JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Second Semester B.E. Degree: Test 1
Department of Mathematics
ENGINEERING MATHEMATICS II

Time : 1 Hour 15 minutes

Date: 9/06/2025

Max. Marks : 30

- NOTE:** 1. Answer all the questions in **PART-A**.
 2. Questions in **PART-B** have internal choice.

PART-A

Q.No.	CO	CD	PI	Question	Marks
1	CO1	Understanding	1.1.1	(i) Solve: $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = 0$ (ii) Solve: $(D^2 + D)y = x^2 + 2x + 4$	08
2	CO2	Remembering	1.1.1	Find Laplace transform of the following: (i) t^2e^{2t} (ii) $(1 + e^t)^2$	07

PART-B

Q.No.	CO	CD	PI	Question	Marks
3 a)	CO1	Understanding	1.2.1	Solve $\frac{d^2y}{dx^2} + y = \tan x$ by the method of variation of parameters.	7
OR					
3 b)	CO1	Understanding	1.2.1	Solve the Cauchy's homogeneous linear equation $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = \log x$.	7

Q.No.	CO	CD	PI	Question	Marks
4 a)	CO2	Remembering	1.2.1	If $f(t)$ is a periodic function with the period T , then prove that $L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$.	8
OR					
4 b)	CO2	Remembering	1.1.1	(i) Prove that $\int_0^\infty e^{-t} \frac{\sin t}{t} dt = \frac{\pi}{4}$. (ii) Find Laplace transform of the function $t^2u(t-3)$.	8

Course Outcome: At the end of the course the students will have the ability to

CO 1	Identify second and higher order differential equations and apply to engineering problems such as oscillation of simple pendulum spring and LCR circuits.
CO 2	Apply the concept of Laplace transform and obtain Laplace transform of periodic functions and unit step functions to solve the problems in transforming the continuous signals.

Performance Indicator :

1.1.1	Apply mathematical techniques such as calculus, linear algebra and statistics to solve problems.
1.2.2	Apply laws of natural science to engineering problems.